

Experiment 6: Breadth First Search (BFS)

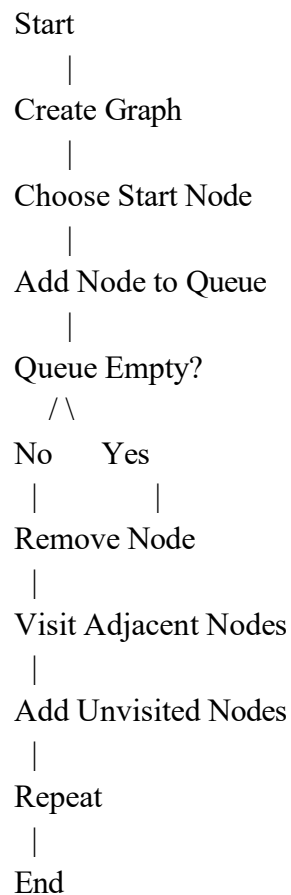
Aim

To implement the Breadth First Search (BFS) algorithm in Python for graph traversal.

Objective

- To traverse all vertices level by level.
- To find the shortest path in an unweighted graph.

Flowchart



Algorithm

1. Create a graph using an adjacency list.
2. Select the starting node.
3. Add the starting node to the queue.
4. Mark it as visited.
5. Remove a node from the queue.
6. Visit all unvisited adjacent nodes.
7. Add them to the queue.
8. Repeat until the queue becomes empty.

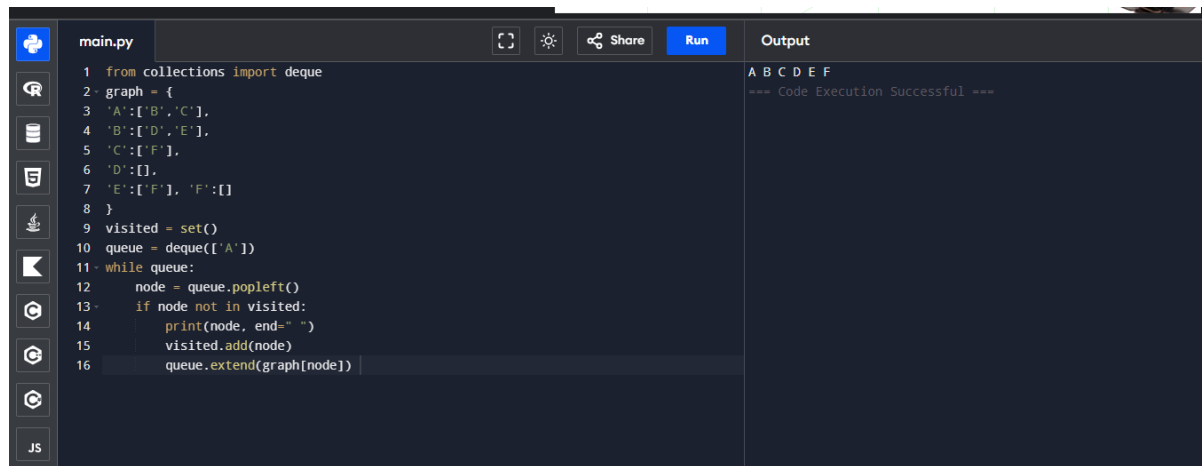
Program

```
from collections import deque graph = {  
    'A':['B','C'],  
    'B':['D','E'],  
    'C':['F'],  
    'D':[],  
    'E':['F'], 'F':[]  
}
```

```
visited = set() queue =  
deque(['A'])
```

```
while queue:  
    node = queue.popleft() if node  
    not in visited:  
        print(node, end=" ")  
        visited.add(node)  
        queue.extend(graph[node])
```

Output



The screenshot shows a code editor with a file named `main.py`. The code implements a Breadth First Search (BFS) algorithm on a graph. The graph is defined as follows:

- `A` is connected to `B` and `C`.
- `B` is connected to `D` and `E`.
- `C` is connected to `F`.
- `D` has no outgoing edges.
- `E` is connected to `F`.
- `F` has no outgoing edges.

The algorithm starts at node `A`, marks it as visited, and adds it to a queue. It then processes the queue level by level, printing each node as it is visited. The output shows the nodes `A B C D E F` in order, followed by the message `=== Code Execution Successful ===`.

```
1 from collections import deque
2 graph = {
3     'A': ['B', 'C'],
4     'B': ['D', 'E'],
5     'C': ['F'],
6     'D': [],
7     'E': ['F'], 'F': []
8 }
9 visited = set()
10 queue = deque(['A'])
11 while queue:
12     node = queue.popleft()
13     if node not in visited:
14         print(node, end=" ")
15         visited.add(node)
16         queue.extend(graph[node])
```

Result

The BFS traversal was performed successfully.

Conclusion

The Breadth First Search algorithm visits nodes level by level using a queue.

